

Akshera Paladhi

apaladhi@ucsc.edu | 510-241-8963 | [linkedin/akshera](https://www.linkedin.com/in/akshera) | aksherap.com

EDUCATION

University of California, Santa Cruz

Bachelor of Science in Electrical Engineering, Minor in Astrophysics

Expected Graduation: June 2028

3.58 GPA

- Relevant coursework:** Electronic Circuits, Introduction to Electricity Industry, Computer Systems and C Programming, Computer Systems and Assembly Language, Scientific Computing, Introduction to Physics I: Mechanics, Introduction to Electricity and Magnetism, Waves, Optics, Thermodynamics, Statics, Dynamics, Linear Algebra, Differential Equations

SKILLS AND INTERESTS

- Software, Languages & Simulation:** C, C++, Python, Verilog/SystemVerilog, VHDL, SQL, Bash/Shell, MATLAB, Linux, Git, NVIDIA Isaac, MuJoCo, STM32CubeIDE, Vivado IDE, Arduino IDE
- Hardware, Tools & Instrumentation:** Altium Designer & KiCad (PCB Design), LTSpice, Siemens NX & SolidWorks & Onshape (CAD), FDM/SLA 3D Printing, STM32, PIC32, ESP32, USB, USB-PD, Type-C, I2C, SPI, UART, RP2040, Oscilloscopes, Multimeters, Power Supplies, Protocol Analyzers, Precision Soldering, V&V Testing
- Certifications & Licenses:** FAA Part 107 Remote Pilot Certification; Certificate of Completion NPWEE Summer 2025 L'SPACE (August 2025)

LEADERSHIP AND RESEARCH EXPERIENCE

CITRIS Drone Initiative for Drone Education and Research Internship

Santa Cruz, CA

Drone Researcher

Jun 2026 – Present

- Designed airframes and mounts in Onshape and SolidWorks, optimizing CAD models for rapid 3D printing (Bambu Lab, PrusaSlicer).
- Assembled, wired, and soldered flight hardware and power distribution systems for custom multicopter testbeds.
- Performed board soldering, hardware rework, and FAA Part 107 flight operations to maintain research drones.

UCSC Institute of Particle Physics (SCIPP)

Santa Cruz, CA

Undergraduate Researcher

Jun 2025 – Present

- Applying analog circuit analysis to simulate and continuously improve the robustness of the high-speed front-end electronics using LTSpice for the Diamond Submerged Wireless Imaging Monitor (SWIM) project.
- Designing a signal-shaping circuit by applying schematic design and debugging skills to stabilize ultra-short detector pulses for proton beam diagnostics at Los Alamos National Lab (LANL), ensuring reliable performance.
- Developing C firmware on an Arduino Nano RP2040 Connect for real-time data acquisition to monitor particle beam profiles and resolve hardware-software integration issues.
- Leveraged Python scripting in Linux to automate real-time testing of Thorlabs cameras, evaluating sensor response and data integrity across various wavelengths.

UC Santa Cruz Quantum Computing

Santa Cruz, CA

Vice President & Researcher

Aug 2025 – Present

- Leading a 10+ member research group in the study of Position-Based Quantum Cryptography (PBQC) utilizing strong interpersonal and organizational skills, applying analytical models to develop secure authentication frameworks.
- Collaborating with 3+ UC campuses and the Quantum Coalition to expand shared research resources.

Learning Support Services

Santa Cruz, CA

Engineering Tutor

Sep 2025 – Present

- Mentoring 15+ students weekly by simplifying concepts in computer systems, digital logic, CMOS gates, transistors, Boolean algebra, and RISC-V architecture into practical problem-solving strategies.

TECHNICAL PROJECTS

Interactive Household Robotics (Lamp & Alarm Clock)

Jun 2026 – Present

- Designed multi-DOF robot CAD in Onshape and SolidWorks, optimizing dimensions and parts for DFM/DFA to cut total assembly components by 25%.
- Hand-assembled articulated electromechanical linkages, integrating custom fasteners, adhesives, routing light wiring, and validating connector fits.
- Designed and fabricated custom benchtop test fixtures to validate stability, acoustics, FEA, and kinematic motion profiles across dynamic load states.
- Routed custom 2-layer PCBs in KiCad integrating switching regulators, LDOs, and DC-DC converters, executing power distribution debugging to ensure stable multi-actuator operation.

Air Mouse (ESP32 + MPU-6050)

Dec 2025 – Jan 2026

- Developing a wireless motion-controlled robotics interface using an ESP32 microcontroller and MPU-6050 IMU.
- Implemented sensor fusion and custom filtering algorithms in C to translate real-time inertial data, cutting Bluetooth HID transmission latency by 30%.
- Designed a custom PCB in KiCad and executing precision SMD soldering to deliver production-ready hardware.

Embedded Systems BattleBoats Game

May 2025 – June 2025

- Built a two-player game in Embedded C using a custom UART protocol to sync real-time game data between microcontroller nodes.
- Designed a synchronous Finite State Machine with packet framing and hashed validation keys, reducing packet transmission errors by 95%.
- Debugged circuits with oscilloscopes and logic analyzers to capture serial streams and resolve timing jitter.